

The Case of the Missing Compass

Someone took the demonstration compass off Mr. Nappier's desk during 4th period. You have five suspects, a pile of conditional statements, and one afternoon. Logic only — no guessing.

Part A — Truth or Trap (2.1: converse, inverse, contrapositive)

Each witness statement is a TRUE conditional. For each one, write the converse, inverse, and contrapositive, and mark each T or F. A FALSE statement is a trap — the detective who believes it will accuse the wrong person.

1. "If a suspect was in Room 406 during 4th period, then the suspect had access to the compass."
2. "If Preston was at the dentist, then Preston was not in Room 406."
3. The detective's rule: of the three rewrites, which one is ALWAYS as true as the original? Write it in one sentence, and say why the other two are traps.

Part B — Counterexample Hunt (2.2: conjectures)

A junior detective wrote these conjectures from a few observations. Bust each one with a counterexample — or, if it can't be busted, write "no counterexample" and say what kind of reasoning would be needed to PROVE it.

4. "If a number is odd, then it is prime."
5. "If two angles are congruent, then they are vertical angles."
6. "If $x^2 = 49$, then $x = 7$."
7. "The sum of two even numbers is even."

Part C — Sherlock Was Wrong (2.2: inductive vs. deductive)

Label each piece of reasoning INDUCTIVE (a pattern from examples) or DEDUCTIVE (a certain conclusion from accepted facts and logic).

8. "Every time Marcus has shavings on his sleeve, he has just sharpened a pencil. He has shavings now, so he just sharpened a pencil."
9. "All students in 4th-period geometry took the quiz. Jaden is in 4th-period geometry. So Jaden took the quiz."
10. "Sherlock Holmes looks at a stranger's tan and posture and announces, 'You have been in Afghanistan.'"
11. "If it's a square, it's a rectangle. If it's a rectangle, it has four right angles. This is a square, so it has four right angles."

Part D — Detective Deduction (2.2: Law of Detachment, Law of Syllogism)

The clues below are all TRUE. Chain them with the Law of Syllogism, plug in facts with the Law of Detachment, and find the culprit. Warning: one clue is a converse in disguise — using it backwards is the trap. Show every step and name the law you used.

#	Clue
C1	If a suspect took the compass, then the suspect was in Room 406 during 4th period.
C2	If a suspect was in Room 406 during 4th period, then the suspect is in Mr. Nappier's 4th-period class.
C3	If a suspect is in Mr. Nappier's 4th-period class, then the suspect had a geometry quiz that day.
C4	If a suspect had a geometry quiz that day, then the suspect had pencil shavings on their sleeve.
C5	If a suspect has a locker on the second floor, then the suspect walks past Room 406 every day.

Suspect	Facts
Preston	At the dentist all of 4th period (signed note).
Marcus	Pencil shavings on his sleeve. Locker on the second floor. Not in any 4th-period geometry class.
Jaden	In Mr. Nappier's 4th-period class. Pencil shavings on his sleeve. Was in Room 406 during 4th period.
Theo	Locker on the second floor. Walks past 406 every day. In 6th-period geometry, not 4th.
Liam	Pencil shavings on his sleeve. Had a geometry quiz that day — in Mrs. Sierra's class, Room 210.

- 12.** Chain $C1 \rightarrow C2 \rightarrow C3 \rightarrow C4$ with the Law of Syllogism into ONE conditional: "If a suspect took the compass, then _____."
- 13.** Marcus, Jaden, and Liam all have pencil shavings. Does the chained statement in #12 let you conclude that any of them took the compass? Explain using the word CONVERSE.
- 14.** Now use the CONTRAPOSITIVE of C1. Which suspects can you ELIMINATE, and by what law?
- 15.** Who is left? Does the logic PROVE that person took the compass, or only that they COULD have? Be precise.
- 16.** Clue C5 was never useful. Write one sentence explaining why — what would it take for C5 to help?